

Evaluation of cholestasis in Iranian infants less than three months of age.

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The aim of this study is to find-out the possible etiologies in Iranian infants less than three months in Shiraz, South of Iran. Cholestatic jaundice most probably occurs due to a pathological condition and the most frequent causes in early infancy are neonatal hepatitis and biliary atresia. Early diagnosis and treatment of infantile cholestasis can improve prognosis of liver diseases by prevention of the complications of these disorders. In this retrospective study, 122 infants under 3 months of age with cholestasis were studied in Nemazee Hospital (affiliated to Shiraz University of Medical Sciences) during the years 2001-2011. Demographic data, duration of jaundice, liver biopsy and the causes of cholestasis were recorded. There were 76 males (62.3%) and 46 females (37.7%) with a mean age of 54.4 ± 23.7 days. The most common clinical finding was jaundice that was seen in all patients (100%). The onset of jaundice was the first day to the fifty two days of age, with an average age of 15.6 ± 16.1 days. Other findings included hepatomegaly in 92 patients (76.4%), clay-color stool in 54 (44.3%), and splenomegaly in 29 patients (23.8%). In this study, the most common causes of cholestasis were biliary atresia (30=24.6%), idiopathic neonatal hepatitis (30= 24.6%) and bile ducts paucity (16=10.3). The results of this study showed that biliary atresia and neonatal hepatitis are the most common causes of infantile cholestasis in this area. It is recommended that biliary atresia should be discriminated from other forms of neonatal cholestasis.

The origin and cell lineage of microglia: new concepts.

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Despite intense study, the precise origin and cell lineage of microglia, the resident mononuclear phagocytes of the nervous system, are still a matter for debate. Unlike macroglia (astrocytes and oligodendrocytes) and neurons, which are derived from neuroectoderm, microglial progenitors arise from peripheral mesodermal (myeloid) tissue. The view still commonly held is that tissue-resident mononuclear phagocytes (including microglia) are derived from circulating blood monocytes and these take up residence late in gestation and postnatally. However, microglial progenitors colonise the nervous system primarily during embryonic and fetal periods of development. Recent evidence indicates differences between the lineage of mononuclear phagocytes during the embryonic and fetal period from that in the neonate and adult-mononuclear phagocytes that take up residence within tissues are derived from a lineage of myeloid cells that is independent of the monocyte lineage. Our own findings on the development and differentiation of microglial progenitors, taken together with findings by other investigators, and in the context of the heterogeneity between myeloid differentiation in the fetus and in the adult, support the view that microglia are derived prenatally from mesodermal progenitors that are distinct from monocytes. Furthermore, microglial progenitors colonise the nervous system via extravascular routes initially. These findings challenge the concept that resident microglia in the nervous system are derived from circulating blood monocytes. Work is still underway to establish the tissue of origin and lineage of microglial progenitors in vivo. This information is critical not only from a developmental perspective, but significantly from a therapeutic viewpoint, as (i) the unique property of microglial progenitors to colonise the nervous system from the periphery allows these cells to be exploited as a biological and non-invasive means for cell therapy by delivering genes to the nervous system (microglial engraftment), and (ii) there are indications that microglial progenitors are specifically able to home to the nervous system. Use of microglial progenitors for therapeutic purposes becomes feasible only if the origin and cell lineage of these microglial progenitors are known and these cells can

be isolated and manipulated in vitro (i.e., to express specific trophic factors) prior to therapeutic transfer (e.g., intravenously) in vivo. In this paper, we shall briefly consider the existing concepts on the origin and lineage of microglial progenitors and discuss new hypotheses in the light of emerging data that suggest clear differences between fetal and adult ontogeny of myeloid cells.

Fate mapping analysis reveals that adult microglia derive from primitive macrophages.

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Microglia are the resident macrophages of the central nervous system and are associated with the pathogenesis of many neurodegenerative and brain inflammatory diseases; however, the origin of adult microglia remains controversial. We show that postnatal hematopoietic progenitors do not significantly contribute to microglia homeostasis in the adult brain. In contrast to many macrophage populations, we show that microglia develop in mice that lack colony stimulating factor-1 (CSF-1) but are absent in CSF-1 receptor-deficient mice. In vivo lineage tracing studies established that adult microglia derive from primitive myeloid progenitors that arise before embryonic day 8. These results identify microglia as an ontogenically distinct population in the mononuclear phagocyte system and have implications for the use of embryonically derived microglial progenitors for the treatment of various brain disorders.

Modulation of Hematopoietic Lineage Specification Impacts TREM2 Expression in Microglia-Like Cells Derived From Human Stem Cells.

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Microglia are the primary innate immune cell type in the brain, and their dysfunction has been linked to a variety of central nervous system disorders. Human microglia are extraordinarily difficult to obtain for experimental investigation, limiting our ability to study the impact of human genetic variants on microglia functions. Previous studies have reported that microglia-like cells can be derived from human monocytes or pluripotent stem cells. Here, we describe a reproducible relatively simple method for generating microglia-like cells by first deriving embryoid body mesoderm followed by exposure to microglia relevant cytokines. Our approach is based on recent studies demonstrating that microglia originate from primitive yolk sac mesoderm distinct from peripheral macrophages that arise during definitive hematopoiesis. We hypothesized that functional microglia could be derived from human stem cells by employing BMP-4 mesodermal specification followed by exposure to microglia-relevant cytokines, M-CSF, GM-CSF, IL-34, and TGF- β . Using immunofluorescence microscopy, flow cytometry, and reverse transcription polymerase chain reaction, we observed cells with microglia morphology expressing a repertoire of markers associated with microglia: Iba1, CX3CR1, CD11b, TREM2, HexB, and P2RY12. These microglia-like cells maintain myeloid functional phenotypes including A β peptide phagocytosis and induction of pro-inflammatory gene expression in response to lipopolysaccharide stimulation. Addition of small molecules BIO and SB431542, previously demonstrated to drive definitive hematopoiesis, resulted in decreased surface expression of TREM2. Together, these data suggest that mesodermal lineage specification followed by cytokine exposure produces microglia-like cells in vitro from human pluripotent stem cells and that this phenotype can be modulated by factors influencing hematopoietic lineage in vitro.

Brain-Engrafted Monocyte-derived Macrophages from Blood and Skull-Bone Marrow Exhibit Distinct Identities from Microglia.

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Microglia are thought to originate exclusively from primitive macrophage progenitors in the yolk sac (YS) and to persist throughout life without much contribution from definitive hematopoiesis. Here, using lineage tracing, pharmacological manipulation, and RNA-sequencing, we elucidated the presence and characteristics of monocyte-derived macrophages (MDMs) in the brain parenchyma at baseline and during microglia repopulation, and defined the core transcriptional signatures of brain-engrafted MDMs. Lineage tracing mouse models revealed that MDMs transiently express CD206 during brain engraftment as CD206 + microglia precursors in the YS. We found that brain-engrafted MDMs exhibit transcriptional and epigenetic characteristics akin to meningeal macrophages, likely due to environmental imprinting within the meningeal space. Utilizing parabiosis and skull transplantation, we demonstrated that monocytes from both peripheral blood and skull bone marrow can repopulate microglia-depleted brains. Our results reveal the heterogeneous origins and cellular dynamics of brain parenchymal macrophages at baseline and in models of microglia depletion.

Convergence between Microglia and Peripheral Macrophages Phenotype during Development and Neuroinflammation.

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Differently from other myeloid cells, microglia derive exclusively from precursors originating within the yolk sac and migrate to the CNS under development, without any contribution from fetal liver or postnatal hematopoiesis. Consistent with their unique ontology, microglia may express specific physiological markers, which have been partly described in recent years. Here we wondered whether profiles distinguishing microglia from peripheral macrophages vary with age and under pathology. To this goal, we profiled transcriptomes of microglia throughout the lifespan and included a parallel comparison with peripheral macrophages under physiological and neuroinflammatory settings using age- and sex-matched wild-type and bone marrow chimera mouse models. This comprehensive approach demonstrated that the phenotypic differentiation between microglia and peripheral macrophages is age-dependent and that peripheral macrophages do express some of the most commonly described microglia-specific markers early during development, such as *Fcrls*, *P2ry12*, *Tmem119*, and *Trem2*. Further, during chronic neuroinflammation CNS-infiltrating macrophages and not peripheral myeloid cells acquire microglial markers, indicating that the CNS niche may instruct peripheral myeloid cells to gain the phenotype and, presumably, the function of the microglia cell. In conclusion, our data provide further evidence about the plasticity of the myeloid cell and suggest caution in the strict definition and application of microglia-specific markers. SIGNIFICANCE STATEMENT Understanding the respective role of microglia and infiltrating monocytes in neuroinflammatory conditions has recently seemed possible by the identification of a specific microglia signature. Here instead we provide evidence that peripheral macrophages may express some of the most commonly described microglia markers at some developmental stages or pathological conditions, in particular during chronic neuroinflammation. Further, our data support the hypothesis about phenotypic plasticity and convergence among distinct myeloid cells so that they may act as a functional unit rather than as different entities, boosting their mutual functions in different phases of disease. This holds relevant implications in the view of the growing use of myeloid cell therapies to treat brain disease in humans.

Origin and Emergence of Microglia in the CNS-An Interesting (Hi)story of an Eccentric Cell.

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Microglia belong to tissue-resident macrophages of the central nervous system (CNS), representing the primary innate immune cells. This cell type constitutes ~7% of non-neuronal cells in the mammalian brain and has a variety of biological roles integral to homeostasis and pathophysiology from the late embryonic to adult brain. Its unique identity that distinguishes its "glial" features from tissue-resident macrophages resides in the fact that once entering the CNS, it is perennially exposed to a unique environment following the formation of the blood-brain barrier. Additionally, tissue-resident macrophage progenies derive from various peripheral sites that exhibit hematopoietic potential, and this has resulted in interpretation issues surrounding their origin. Intensive research endeavors have intended to track microglial progenitors during development and disease. The current review provides a corpus of recent evidence in an attempt to disentangle the birthplace of microglia from the progenitor state and underlies the molecular elements that drive microgliogenesis. Furthermore, it caters towards tracking the lineage spatiotemporally during embryonic development and outlining microglial repopulation in the mature CNS. This collection of data can potentially shed light on the therapeutic potential of microglia for CNS perturbations across various levels of severity.

Peripheral monocyte-derived cells counter amyloid plaque pathogenesis in a mouse model of Alzheimer's disease.

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Microglia, the parenchymal tissue macrophages in the brain, surround amyloid plaques in brains of individuals with Alzheimer's disease (AD) but are ineffective at clearing amyloid to mitigate disease progression. Recent studies in mice indicate that microglia are derived exclusively from primitive yolk sac hematopoiesis and self-renew without contribution from ontogenically distinct monocytes/macrophages of definitive adult hematopoietic origin. Using a genetic fate-mapping approach to label cells of definitive hematopoietic origin throughout life span, we discovered that circulating monocytes contribute 6% of plaque-associated macrophages in aged AD mice. Moreover, peripheral monocytes contributed to a higher fraction of macrophages in the choroid plexus, meninges, and perivascular spaces of aged AD mice versus WT control mice, indicating enrichment at potential sites for entry into the brain parenchyma. Splenectomy, which markedly reduced circulating Ly6Chi monocytes, also reduced abundance of plaque-associated macrophages of definitive hematopoietic origin, resulting in increased amyloid plaque load. Together, these results indicate that peripherally derived monocytes invade the brain parenchyma, targeting amyloid plaques to reduce plaque load.

MafB antagonizes phenotypic alteration induced by GM-CSF in microglia.

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Microglia are tissue-resident macrophages which are distributed throughout the central nervous system

(CNS). Recent studies suggest that microglia are a unique myeloid population distinct from peripheral macrophages in terms of origin and gene expression signature. Granulocyte-macrophage colony-stimulating factor (GM-CSF), a pleiotropic cytokine regulating myeloid development, has been shown to stimulate proliferation and alter phenotype of microglia in vitro. However, how its signaling is modulated in microglia is poorly characterized. MafB, a bZip transcriptional factor, is highly expressed in monocyte-macrophage lineage cells including microglia, although its role in microglia is largely unknown. We investigated the crosstalk between GM-CSF signaling and MafB by analyzing primary microglia. We found that *Mafb*-deficient microglia grew more rapidly than wild-type microglia in response to GM-CSF. Moreover, the expression of genes associated with microglial differentiation was more downregulated in *Mafb*-deficient microglia cultured with GM-CSF. Notably, such differences between the genotypes were not observed in the presence of M-CSF. In addition, we found that *Mafb*-deficient microglia cultured with GM-CSF barely extended their membrane protrusions, probably due to abnormal activation of RhoA, a key regulator of cytoskeletal remodeling. Altogether, our study reveals that MafB is a negative regulator of GM-CSF signaling in microglia. These findings could provide new insight into the modulation of cytokine signaling by transcription factors in microglia.

Neuronal extracellular microRNAs miR-124 and miR-9 mediate cell-cell communication between neurons and microglia.

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In contrast to peripheral macrophages, microglia in the central nervous system (CNS) exhibit a specific deactivated phenotype; however, it is not clear how this phenotype is maintained. Two alternative hypotheses were postulated recently: (a) microglia differ from peripheral macrophages being derived from the yolk sac (YS), whereas peripheral macrophages originate from bone marrow (BM); (b) microglia acquire a specific phenotype under the influence of the CNS microenvironment. We have previously shown that microglia express miR-124, which was also induced in BM-derived macrophages co-cultured with a neurons. We here investigated the possibility of horizontal transfer of the neuron-specific microRNAs miR-124 and miR-9 from primary neurons to microglia/macrophages. We found that after incubation with neuronal conditioned media (NCM), macrophages downregulated activation markers MHC class II and CD45. Neither cultured adult microglia nor YS- and BM-derived macrophages demonstrated intrinsic levels of miR-124 expression. However, after incubation with NCM, miR-124 was induced in both YS- and BM-derived macrophages. Biochemical analysis demonstrated that the NCM contained miR-124 and miR-9 in complex with small proteins, large high-density lipoproteins (HDLs), and exosomes. MiR-124 and miR-9 were promptly released from neurons, and this process was inhibited by tetrodotoxin, indicating an important role of neuronal electric activity in secretion of these microRNAs. Incubation of macrophages with exogenous miR-124 resulted in efficient translocation of miR-124 into the cytoplasm. This study demonstrates an important role of neuronal miRNAs in communication of neurons with microglia, which favors the hypothesis that microglia acquire a specific phenotype under the influence of the CNS microenvironment.